

LYTRONTM MCS

MODULAR COOLING SYSTEMS

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***Installation and Operational
Instructions
FOR
Lytron Model 684864
Cooling System***

INSTALLATION AND OPERATIONAL INSTRUCTIONS

LYTRON MODEL 6848G4 COOLING SYSTEM

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PREFACE TO INSTRUCTIONS

The Lytron Model 684864 Cooling System has been approved for use with Medical Equipment by meeting requirements as specified in the Unified UL, CSA and IEC Safety Standards for Medical Electrical Equipment.

It has been specified, designed and approved for Operation and Storage in a controlled environment as follows:

- Operation is to be in a Machinery Room (or its equivalent Ancillary Support Equipment Room), which is completely separate from the room in which Medical Patients are being treated by Medical Personnel.
- During continuous duty operation, access to this Machinery or Equipment Room will be normally by Operation and Maintenance Personnel ONLY.
- Both Operation and Storage will be in rooms that are inside of a building, protected from outside environmental weather conditions.
- At all times, Storage and Operational Rooms are to be with temperatures controlled within the range of 60 - 80 (°F), equivalent to 15.6 - 26.7 (°C).

This Equipment has been approved as TYPE B EQUIPMENT, having been designed for reliable service on a CONTINUOUS OPERATIONAL DUTY CYCLE.

INSTALLATION AND OPERATIONAL INSTRUCTIONS LYTRON MODEL 6848G4 COOLING SYSTEM

STEP 1: REMOVE FROM SHIPPING CONTAINER.

Using the finger-grips on each side of the system cover, lift the 684864 Cooling System straight up and out of the shipping container. Remove any protective styrofoam or cardboard.

STEP 2: COOLANT LINE INSTALLATION.

Length of runs should be as short as possible to reduce pressure drop losses. Lines are best *run* at or near the same level as the cooling system, until reaching the equipment to be cooled.

Both supply and return lines (which deliver coolant to and from the equipment being cooled) should be nominally 1/2" ID hose. The COOLANT OUT and COOLANT IN ports are both located on the back of the 684864; and both connections are with hose-barb fittings for 1/2" ID hose.

STEP 3: COOLING AIR FLOW INSTRUCTIONS.

Avoid potential air flow obstructions to the 684864 Cooling System. Make sure that there are no obstructions to air flow from under the enclosure openings for at least 6" along both sides and the back of the unit.

STEP 4: CONNECTING ELECTRICAL POWER.

IMPORTANT: *Before connecting electrical power, the external potential equalization conductor wire must be secured to the 1/4-20 threaded terminal stud located on the back of the unit. The knurled thumbscrew and split lockwasher (shipped with unit) are first unscrewed **from** the threaded terminal stud (DO NOT USE PLIERS).*

Place the equalization conductor lug onto the terminal stud. Now replace the split lock washer and knurled thumbscrew back onto the terminal stud (in that order), and tighten down the thumbscrew (BY HAND ONLY, NO TOOLS).

Make sure the cooling system power inlet cord and plug set have not yet been energized.

Connect the power cord and plug into the back of the 684864 Cooling System. **NOTE: IF PUMP AND FAN MOTORS START, IMMEDIATELY DISCONNECT POWER PLUG.**

Recheck to make sure the power inlet cord and plug set have not yet been energized. **RUNNING THE SYSTEM DRY FOR EVEN A SHORT TIME WILL DAMAGE THE PUMP.** Now repeat the previous step until the power plug has been connected, without either the pump or fan motor running.

STEP 5: INITIAL FILLING OF RESERVOIR.

Fill the reservoir to the coolant FULL mark on the front panel.

STEP 6: PRIMING PUMP.

It is now safe to apply power and to quickly "jog" the pump "ON-and-OFF". The fan will also turn "on-and-off" simultaneously. This is normal. We are now "priming" the pump. Continue to do this several times, while watching the coolant level indicator on the front panel until the coolant level in the reservoir has dropped down to the coolant **ADD** mark on the front panel. Now turn off the system.

STEP 7: FILLING SYSTEM.

Refill the reservoir to the FULL mark on the front panel.

Now repeat this "jogging" and refilling process until the system is totally full and coolant begins to return back into the reservoir. This will be confirmed when the decrease in coolant level begins to slow down. The coolant level will eventually become stable when all the liquid passages have become full, and as much coolant is being pumped out of the reservoir as is being returned to it.

The reason for the slow filling procedure is to slowly push the air out of the coolant passages, back into the reservoir and out the filler opening. The slower the better, so that air won't become "entrained", recirculating with the coolant and causing it to become "frothy".

STEP 8: "DRAIN-DOWN".

After the level has become stable (while still continuously jogging on and off), turn the pump off and observe the coolant level. If the coolant level remains the same, all is well. **HOWEVER, IF THE LEVEL SLOWLY BEGINS TO RISE, IT MEANS COOLANT AT A HIGHER LEVEL IS DRAINING BACK DOWN INTO THE RESERVOIR.**

In this event, turn the pump back on again to return the tank to a steady level. Raise the 684864 to a higher position, to see if this will temporarily correct the problem.

This "drain-down" problem can be permanently corrected by (1) either installing a shutoff valve in the return line (either manual or automatic); or (2) by using a non-vented cap on the reservoir (during shutdown only).

STEP 9: "ON-LINE".

Your 684864 Cooling System is now ready for full-time use. **It has been designed for reliable service on a CONTINUOUS OPERATIONAL DUTY CYCLE.**

SYSTEM MAINTENANCE

(A) Scheduled Maintenance

Routine system maintenance should be very minimal once the 684864 Cooling System has been initially installed, filled, primed and is up and running. Maintenance instructions for changing and maintaining coolant level, not covered in the installation and operating instructions, will be covered here under **SYSTEM MAINTENANCE**.

Operating experience with each unique equipment installation will dictate reasonable frequency of system inspections and/or scheduled maintenance. Suggested guidelines are presented under the following Sections (B), (C) and (E).

(B) Weekly Inspections

Noise Level: Any abnormal sound or substantial increase in noise level since the last weekly inspection may indicate an impending pump, fan or coolant blockage problem which should be further investigated.

Leakage: The weekly visual check does not require removal of the top cover. However, observation of coolant on the floor surface, coming out from under the system, calls for a further check for possible new leaks.

Coolant Level: Any significant drop in the coolant level on the front panel since the previous weekly check should be investigated further. If there is no visual system leak, then the loss may be due to leakage elsewhere in the equipment. See Section (D) for instructions on "topping up" the reservoir.

Pump and Fan Assemblies: With upwards of 10,000 hours continuous duty life expectancy on the pump and 20,000 hours on the fan, both of these components should check OK, on a week-after-week basis.

Coolant Temperature: With no temperature monitor in the system, checking each week by placing a hand on top of the cover, will determine if relative operating temperature level is consistently the same as the week before (normal temp level checks OK). Any substantial increase in temperature level should be further investigated (i.e., air flow obstruction, filter blockage).

Filter: With a clean system, the filter becomes a "Watchdog" filter, to make sure no particulate or foreign matter enters the cooling system. With an unusually dirty system, filter buildup will require the screen to be cleaned or changed (see appendix (E) for replacement parts). If the weekly check detects a noticeable increase in temperature and/or a decrease in system flow (check the reservoir for a decrease in visual flow turbulence), then these are potential signs of filter buildup.

(C) Coolant Life

Monthly Inspections: This EGW coolant is very similar to the "anti-freeze" in an automobile cooling system, which only gets drained and replaced about every year or two. How often depends on type of service. Monthly observation on condition of the coolant in the reservoir will begin to indicate replacement cycle for each particular installation. These checks should usually indicate a clear coolant,

the typical "fresh" EGW aroma, and a clean reservoir. Deterioration of any one of these characteristics requires a more frequent check.

Coolant Replacement Cycle: Once the previous monthly inspections have determined length of time before the EGW coolant must be replaced, a replacement schedule can be established for that particular installation's coolant service cycle. The replacement schedule can now replace the monthly inspection which has established the cycle.

(D) Maintaining Coolant Level

If any of the previous observations require coolant level to be brought back up to normal, procedure is as follows:

Remove cover plate (two 1/4-turn screws) from top of cover; remove cap from reservoir.

Add fresh 20% EGW into the reservoir, up to the FULL mark on the front panel. Replace the cap and cover plate.

(E) Level Switch

Inspection Schedule: The level switch protects the pump in the event of accidental loss of fluid. Since this switch is "passive" during normal operation, it is advisable to "exercise" it about once every 6 months, to make sure it is still functional and ready to protect the pump if necessary.

For a full functional check of both switch and circuitry, the system top cover must be removed; the reservoir drained; proper pump shutoff observed; and the reservoir refilled. See maintenance section (F) for reservoir drain and refill procedures.

If switch fails to shut off the pump as reservoir is drained,

DO NOT LET PUMP CONTINUE TO RUN WITH EMPTY INLET HOSE!!

See Appendix (E) for replacement parts and instructions.

(F) Changing Coolant

Pull Power Plug: The Hubbell Power Plug is located on the back of the system. Facing the back side of the assembly, it is located in the lower right corner. Disconnect this power plug to assure there is no live voltage on the unit while changing coolant.

Remove System Cover: First remove cover attaching hardware. There are (12) #8-32 Phillips Pan Head screws with a flat washer and split locking washer on each screw.

After removing all attaching hardware, grasp the "finger-grip" on each side and lift the cover straight up until it is clear of the system.

Drain Coolant From System: The "sight-glass" tube provides the dual function of also being a drain line.

Remove the reservoir cap; disconnect the top end of the "sight-glass" tube from the small fitting elbow on the top of the reservoir; and drop the open end of this tubing into an open container for draining.

Reconnect the small drain tube to the top of the reservoir after draining. Make sure this tube is still in a good position, lined up with the sighting slot on the front panel.

The rest of the system supply and return hoses; and equipment coolant passages can best be drained by disconnecting the system supply and return barbed connections.

Reconnect Hoses: Reconnect system coolant supply and return hoses, making sure the hose clamps are on tight. Also check any other equipment lines that may have been disconnected for draining.

Reconnect Power Plug: Plug should be reapplied with electrical power to the plug **OFF**. **PUMP WILL BE DAMAGED IF RUN DRY.** If pump starts, quickly check to make sure system power to the unit is **OFF** (fan not running), before proceeding. Also check level switch for "hang-up".

Operational Pump/Level Switch Check: With no coolant yet in the system (reservoir empty), when power is applied, the fan will turn on, but the level switch should prevent the pump from becoming energized.

Coolant Refill: Now perform Steps 5, 6 & 7 of the previous Installation and Operational Instructions for initial refilling procedure.

Restart Cooling System: After the coolant level has become stable for several minutes and has been topped off again to the **FULL** mark, the system can now be turned on again to full operation.

Replace System Cover: Before replacing the system cover, carefully recheck a final time for any new leaks.

Reposition system cover and secure by replacing the 12 screws, flat washers and lock washers previously removed.

(G) Filter

With a new system the filter will quickly accumulate any foreign matter introduced during system set-up, filling or contaminants in the coolant itself. This can lead to a decrease in system performance in a short period of time. It is recommended that the filter be replaced after the first month of operation on new systems to insure that the system runs at maximum capacity. After this initial filter replacement, the weekly inspection routine should be adhered to.

APPENDIX

(A) Warning Labels and Symbols

Your 684864 Cooling System has (2) Warning Labels and (3) Symbols which are summarized as follows:



- (A1) This symbol is to alert operating/maintenance personnel that this system is "Type B Equipment". This defines the level of protection against electrical shocks; level of protection as regards leaking currents; and reliability of protective earth connections that have been designed into this unit.



- (A2) This symbol, (a lightning bolt), is mounted on the terminal board cover to warn operating and maintenance personnel of dangerous voltages (115 VAC) on the terminals underneath the cover.



- (A3) The Protective Earth (ground) symbol is located beside the internal ground terminal stud secured to the rear angle of the assembly frame. It assures positive identification of the internal ground terminal for the convenience of maintenance personnel.



- (A4) The Equipotentiality symbol (potential equalization) is located beside the external ground terminal stud secured to the rear angle of the assembly frame. It assures positive identification of the external ground terminal for the convenience of maintenance personnel.



- (A5) This symbol alerts maintenance personnel to consult the accompanying Installation and Operational Instructions Manual for an explanation of the Warning Symbol next to it. In this case, the High Voltage Warning on the terminal board cover brings about the need for this symbol.

(B) Description, Handling and Disposition of Coolant

Potential Danger: Ethylene Glycol is listed as a **TOXIC MATERIAL** and EGW coolant mixtures as used in this system are defined as **HAZARDOUS**. It is a clear, colorless liquid, with a mild odor.

If swallowed, coolant would be HARMFUL or may even be FATAL!! May cause irritation to eyes, skin and breathing path. If ingested, seek IMMEDIATE medical attention.

Work Practices: Drowsiness, narcosis and unconsciousness are possible upon exposure to high concentrations in poorly ventilated, confined spaces.

Wash skin and clothing after exposure; slightly irritating to skin after prolonged contact. Avoid water Contamination and periods of exposure to high temperature.

Eye protection is advised (chemical goggles). It is believed to cause slight eye irritation.

Fire Protection: When subjected to heat or nearby combustion, carbon monoxide and carbon dioxide may be formed on burning in a limited (unventilated) air supply.

Use water **SPRAY**, dry chemical, foam or carbon dioxide to extinguish. Use water to cool fire exposed containers.

If leak or spill has not ignited, use water **SPRAY** to disperse vapors and to provide protection for persons attempting to stop the leak.

Disposal: Does NOT meet criteria for a hazardous waste IF DISCARDED in its purchased form.

**(C) Operating Conditions
Storage Conditions**

Both Operation and Storage of equipment are in the same controlled environmental conditions, which are as follows:

Range of temperatures: 60-80°F (15.6-26.7°C).

Location: Building interior only, machinery or storage room.

(D) Special Shipping Instructions

Shipping conditions **are not** a controlled environment; temperatures during winter months may well be much lower than the operation and storage conditions as specified. To prevent freeze-up during shipment, the following observations and special instructions should be noted.

From the Factory: The 684864 Cooling system is shipped dry from the factory - no problem here.

After Initial Filling: For any subsequent shipments anytime after the unit has once been filled for operations, the following applies:

- a Normal draining procedures do not drain coolant from the Heat Exchanger; and may not completely drain the pump cavity. Under these conditions, coolant freeze-up during shipment could cause damage to either or both of these components.

Shipments Above 20°F: The specified coolant (mixture of 20% Ethylene Glycol, 80% Water) protects the unit from freeze-up down to approximately 20°F (-6.7°C); no problem for shipments above this temperature.

Shipments Below 20°F: For shipments below 20°F (-6.7°C), the easiest approach is to increase the residual EGW concentration up to 50% for protection during shipment (after the 20% coolant has been drained out). After increasing the concentration, any residual coolant left inside unit passages would now be protected down to -40°F (-40°C). To assure a complete mixing of 50/50 EGW, the following procedure is recommended.

- Drain the Cooling System using normal draining procedures found in Maintenance Section (F). The system reservoir will be drained empty during this procedure.
- Now disconnect the supply and return hoses on the back of the unit and make up a short, temporary hose from the COOLANT OUT port to the COOLANT IN port.
- Fill the reservoir to half-full with Ethylene Glycol and the other half with Water (or fill with a pre-mixed 50/50 EGW mixture).
- Perform Steps 5, 6, and 7 of the Installation Instructions to re-prime the pump and restart system flow through the short external hose loop.
- Run the system for at least one hour minimum, to assure a complete mixing of 50/50 EGW throughout all of the coolant passages. Now drain the unit once again for shipment, using draining procedures found in Maintenance Section (F). Any residual coolant will now not freeze.

(E) Replacement Parts

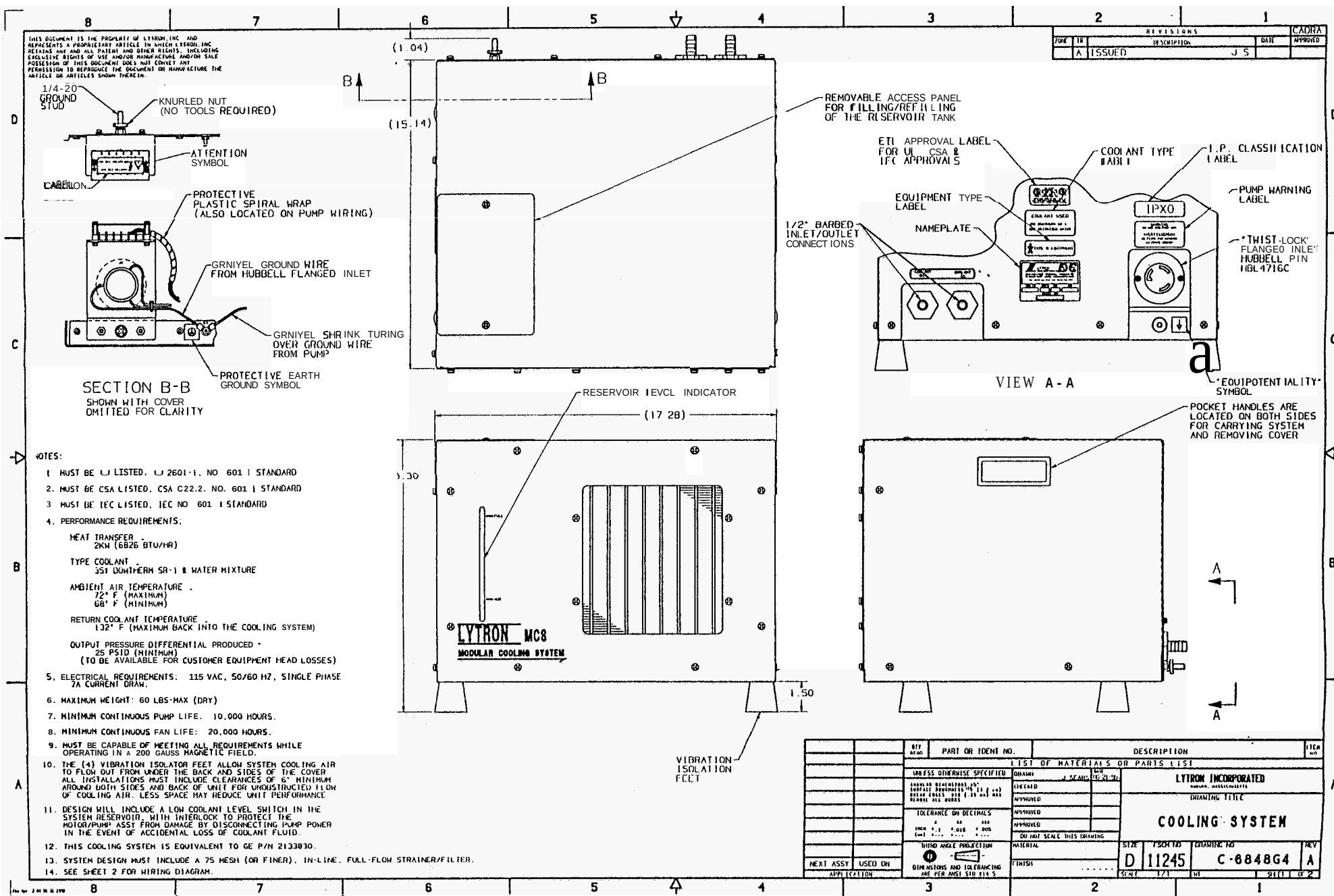
Replacement parts are available in kit form. Each kit is with detailed Installation Instructions which include:

- *Schematic/Wiring Diagram, as may be required.*
- *Part drawing(s) as required to supplement written instructions.*
- *Draining of coolant as may be required.*
- *Removal and replacement of component part.*
- *Refilling of coolant and operational check, **f** required.*
- *Preparation **for** restart, **f** required.*

Kit includes component part plus replacement hardware and electrical parts that might be required.

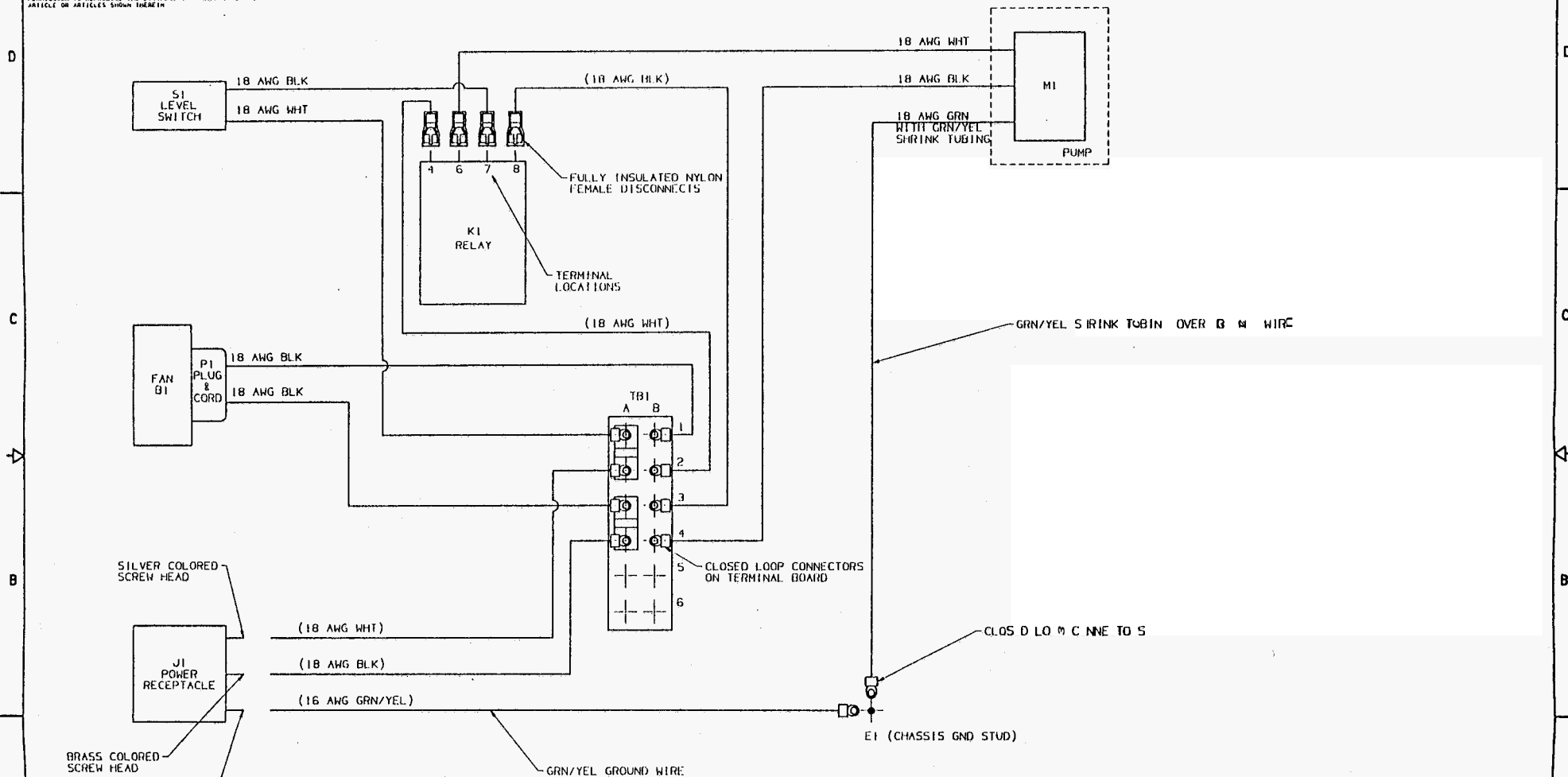
Available replacement parts kits are as follows:

Part Description	Lytron P/N
Motor/Pump Assembly	506840
Heat Exchanger	506841
Fan Assembly	507770
Level Switch	506843
Level Switch Relay	506844
Electrical Power Connector....	506845
Outlet/Inlet Coolant Fitting..	507754
Filter Assembly	506847
Filter Screen	506863
Reservoir Tank Assembly	506864



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